

Warm-up

- (1) Find the Taylor series for e^x centered at $x = 1$.
- (2) Find the Taylor series for $\cos(2x)$ centered at $x = 0$.
- (3) Find the Taylor series for $3 + x - 4x^2$ centered at $x = 0$.

Medium

- (1) Use *the definition of a Taylor series* to find the Taylor series for $3 + x - 4x^2$ centered at $x = 1$.
- (2) Using series we already know, write down a formula for the Taylor series for $x^3 e^{(x^5/4)}$, and then write down the first four non-zero terms of this series.
- (3) Use *the definition of a Taylor series* to find $T_3(x)$ for $\sin(x)$ centered at $x = \pi/3$. (That is, find the terms up through the degree-three term.)
- (4) Using series we already know, write down a formula for the Taylor series and the degree 11 Taylor polynomial $T_{11}(x)$ for $x^2 \ln(1 - 2x^3)$.
- (5) Using series we already know, write down a formula for the (infinite) Taylor series for $e^{3x} - e^x$, and then write down the degree 3 polynomial explicitly.

Stretch

- (1) Write down the formula for $R_k(x) = \cos(x) - T_k(x)$ and find an upper bound on its absolute value. Use this to show that $\cos(x)$ is equal to its Taylor series centered at $x = 0$.
- (2) Find the Taylor series centered at $x = 0$ for the function

$$f(x) = \begin{cases} e^{-1/x^2} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

we first discussed on the 7/6 worksheet.